AI Model Drift, Hallucination & Reliability Monitoring Platform

Project Phase-I Report submitted to Osmania University, Hyderabad, in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering
in
Computer Science

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A.Y. 2026-2027



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CERTIFICATE

This is to certify that the Project Phase-I report entitled **AI Model Hallucination Detection & Response Validation Platform** being submitted by Mr. Mohammad Ahraaz (160923753003), Ms. Ayesha Siddiqua (160923753018), Ms. Soha Mahreen (160923753006) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Computer Science** is a record of bonafide work carried out by them.

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DECLARATION BY THE CANDIDATES

We, **Mr. Mohammad Ahraaz (160923753003), Ms. Ayesha Siddiqua (160923753018) and Ms. Soha Mahreen (160923753006)**, hereby declare that the Project Phase-I report entitled **AI Model Hallucination Detection & Response Validation Platform** is carried out under the guidance of **Mr. Yellaiah Ponnam** Assistant Professor, Department of Information Technology, **Lords Institute of Engineering and Technology**, **Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Computer Science.**

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

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ACKNOWLEDGMENT

We are very pleased to present this thesis of our project. This period of our student life has been truly rewarding a number of people were of immense help to us during the course of our research and the preparation of our thesis.

First, we wish to thank **GOD Almighty** who created heavens and earth, who helped us in completing this project and we also thank our parents who encouraged us in this period of research.

We would like to express our sincere gratitude to **The Management** of **Lords Institute of Engineering and Technology, Hyderabad** for providing us with an excellent platform and resources to successfully carry out this project.

We would also like to thank **Dr. Ravi Kishore Singh, Principal, L.I.E.T.** for extending his help in overall completion of this project work.

We would like to express my deep sense of gratitude **to Dr. K. Nagi Reddy, Professor & HOD**, **Information Technology, Lords Institute of Engineering & Technology, Hyderabad,** for his encouragement and cooperation throughout the project.

We would like to thank **Mr. Yellaiah Ponnam , Assistant Professor, Dept. of Information Technology, Lords Institute of Engineering & Technology**, Hyderabad, our project internal guide, for his guidance and help. His insight during the course of our research and regular guidance were invaluable to us.

Lastly, we acknowledge the support of our faculty members, friends, and family, who inspired and encouraged us throughout this journey.

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DM3: Train the concerned stakeholders on latest technologies to meet the industry needs.

DM4: Inculcate leadership, professional, interpersonal skills and ethical values to address the contemporary societal issues.

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B.E Information Technology Program Educational Objectives (PEOs):

PEO1	Be a competent software engineer in IT and allied industry providing viable solutions.
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B.E Computer Science Program Outcomes (POs):

Engineering Graduates will be able to:

S. No.	Program Outcomes (POs)
1.	**PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2.	**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	**PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	**PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	**PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7.	**PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	**PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9.	**PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	**PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11.	**PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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PSO2	**Problem-Solving Skills:** Deliver quality service to IT industry and provide solutions to real-life problems applying modern computer languages, software methods and tools.

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Course Outcomes: Project Phase- I

After completion of the project, the student will be able to

S.No	Description	Blooms Taxonomy Level
PP.I.CO1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	BTL3
PP.I.CO2	Evaluate different solutions based on economic and technical feasibility	BTL4
PP.I.CO3	Effectively plan a project and confidently perform all aspects of project management.	BTL5
PP.I.CO4	Demonstrate effective written and oral communication skills	BTL4
PP.I.CO5	Prepare the documentation report and perform the presentation of the project work	BTL3

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Course Articulation Matrix:

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2
PP.I.CO1	3	2	2	-	2	2	-	-	3	-	-	2	2
PP.I.CO2	3	3	2	3	2	-	2	-	3	-	-	-	2
PP.I.CO3	3	3	3	-	3	3	1	2	2	3	2	3	-
PP.I.CO4	3	2	2	3	3	3	3	2	-	-	2	3	3
PP.I.CO5	3	3	3	3	3	3	3	2	3	3	-	3	3
AVG	**3**	**2.6**	**2.5**	**3**	**2.6**	**2.7**	**2.2**	**2**	**2.9**	**3**	**2**	**2.7**	**2.5**

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

Level: 1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	✓	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	✓

ABSTRACT

This project proposes an industry-oriented AI Model Hallucination Detection and Response Validation Platform designed to evaluate the reliability and factual consistency of AI-generated responses. The platform analyzes model outputs against available source information, retrieved context, or reference data to identify potentially unsupported, inconsistent, or hallucinated content. It incorporates automated response validation using semantic similarity, contextual grounding, factual consistency, and confidence-based evaluation techniques to determine the reliability of generated responses.The system maintains historical evaluation data to monitor hallucination rates, response quality, model performance, and reliability trends over time. A centralized dashboard provides model health metrics, validation results, alerts, and detailed reports for identifying potential reliability issues. The proposed solution will be implemented as an end-to-end application with APIs, authentication and authorization, data management, logging, testing, and deployment capabilities. The expected outcome is a functional prototype that demonstrates automated AI response evaluation and provides measurable insights into model reliability, creating a foundation for safer and more dependable AI deployment.

Keywords: Artificial Intelligence, Generative AI, Hallucination Detection, Response Validation, Large Language Models (LLMs), AI Reliability

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CHAPTER-1

INTRODUCTION

1.1 Introduction

Artificial Intelligence (AI) has become an essential component of modern software applications and digital services. It enables computer systems to perform tasks such as prediction, classification, recommendation, information retrieval, natural language understanding, and automated decision support. Organizations increasingly use AI-powered solutions in customer support, financial services, healthcare information systems, document processing, fraud detection, predictive maintenance, and enterprise automation.

The development of Machine Learning (ML), Deep Learning, and Large Language Models (LLMs) has significantly expanded the capabilities of intelligent applications. These technologies allow systems to process large volumes of information, identify complex patterns, generate human-like responses, and support activities that previously required considerable human involvement.

However, the successful deployment of an AI model does not guarantee that its performance will remain accurate, consistent, or reliable throughout its operational lifetime. Unlike conventional software applications, AI systems depend heavily on the characteristics of the data they process. Their behavior may change when input distributions shift, user requirements evolve, model configurations are modified, or new versions are introduced.

One important challenge is model drift, which refers to changes in data characteristics or relationships that may affect the behavior of a deployed machine learning model. A model trained using historical information may encounter different conditions after deployment. For example, a fraud detection model developed using previous transaction patterns may become less effective when customer behavior or fraudulent activities change.

Generative AI applications introduce another major concern known as hallucination. Hallucination occurs when an AI model generates information that appears meaningful and convincing but is inaccurate, fabricated, contradictory, or unsupported by available evidence. Large Language Models may produce incorrect factual statements, imaginary

references, misleading explanations, or responses that do not agree with supplied documents.

These issues demonstrate that an AI application can remain technically operational while the quality of its outputs gradually deteriorates. Traditional software monitoring primarily measures infrastructure-related parameters such as CPU utilization, memory consumption, server availability, response time, and application errors. Although these measurements are necessary, they cannot independently determine whether an AI-generated response is factually supported or whether a predictive model continues to perform satisfactorily.

Therefore, AI systems require monitoring mechanisms that evaluate both operational performance and output quality.

The proposed AI Model Drift, Hallucination & Reliability Monitoring Platform aims to address these challenges by providing a centralized environment for evaluating and monitoring different AI models. The platform is designed to collect model execution information, evaluate generated outputs, identify potential hallucinations, detect changes in data distributions, calculate reliability indicators, and present monitoring results through an interactive dashboard.

The system also maintains historical evaluation records, allowing users to compare different model versions, identify unusual changes in performance, and investigate potential reliability issues. Configurable alerts can notify authorized users when selected quality metrics exceed predefined thresholds.

By integrating AI evaluation, drift detection, reliability scoring, monitoring, and reporting into a unified application, the proposed project aims to support the development and maintenance of more dependable AI-powered systems.

1.2 Problem Statement

Organizations increasingly depend on AI models for operational and business decisions, but monitoring their quality after deployment remains challenging. Existing workflows may separate infrastructure monitoring, model evaluation, drift detection, and manual review of generated responses.

This fragmentation makes it difficult to identify emerging quality problems, determine whether a model update introduced a regression, and establish a consistent record of evaluation results.

The major problems are:

1. Model drift: Changes in input distributions or relationships between inputs and outcomes may reduce predictive performance.
2. LLM hallucinations: AI-generated responses may contain unsupported, fabricated, or contradictory information.
3. Incomplete reliability measurement: A single accuracy score does not capture groundedness, consistency, safety, latency, and execution failures.
4. Delayed detection: Quality degradation may remain unnoticed until users report incorrect results.
5. Limited version comparison: Teams may struggle to compare different model, prompt, or configuration versions systematically.
6. Fragmented monitoring: AI-quality metrics and infrastructure metrics may be stored in separate systems.
7. Insufficient investigation evidence: Monitoring alerts may not provide enough information to determine what changed and which model version was affected.
8. Governance challenges: Organizations need traceable evaluation records, documented thresholds, and audit histories.

Therefore, a unified platform is required to monitor AI quality, identify potential drift and hallucinations, calculate reliability indicators, and support systematic investigation of AI-related incidents.

1.3 Objectives of the Project

The primary objective is to develop an integrated platform for monitoring AI model quality and operational reliability.

The specific objectives are:

1. To register AI models, applications, and model versions for monitoring.
2. To collect model execution metadata, including timestamps, latency, status, and token usage where available.
3. To maintain versioned evaluation datasets containing reference answers, expected labels, or supporting evidence.
4. To evaluate model outputs using appropriate quality metrics.
5. To identify potential hallucinations and measure groundedness for supported LLM applications.
6. To detect changes in input features, prediction distributions, and quality metrics.
7. To compare model versions and prompt configurations using consistent evaluation datasets.
8. To calculate configurable reliability scores using selected quality and operational indicators.
9. To generate alerts when monitoring policies are violated.
10. To provide dashboards for quality trends, drift indicators, reliability scores, and incidents.
11. To maintain evaluation history and audit records for traceability.
12. To evaluate the system using controlled datasets, simulated drift, and deliberately constructed unreliable responses.

1.4 Scope of the Project

The project focuses on monitoring AI models after deployment and evaluating candidate models or configurations before release.

1.4.1 Functional scope

The proposed platform includes:

- AI model registry.
- Model and prompt version tracking.
- Telemetry collection.
- Golden evaluation dataset management.
- Automated model-quality evaluation.

- Hallucination and groundedness assessment.
- Data and prediction drift detection.
- Reliability scoring.
- Anomaly detection.
- Threshold-based alerting.
- Model-version comparison.
- Incident management.
- Monitoring dashboards and reports.
- Audit logging.

1.4.2 AI and machine learning scope

For conventional machine learning models, the platform can monitor accuracy, precision, recall, F1-score, input-feature distributions, prediction distributions, and performance changes when labeled outcomes are available.

For LLM-powered applications, it can monitor groundedness, factual consistency with supplied evidence, relevance, instruction adherence, safety criteria, latency, token usage, and execution failures.

The evaluation methods are selected according to the model type and application requirements. Hallucination detection is treated as an evaluation problem with uncertainty rather than a guarantee of perfect factual verification.

1.4.3 Data scope

The platform may process:

- Model identifiers and versions.
- Prompt and configuration versions.
- Model inputs and generated outputs, where permitted.
- Reference questions and expected answers.
- Supporting documents or retrieved context.
- Evaluation metrics.
- Latency and execution status.
- Drift statistics.
- Alert and incident records.

Synthetic telemetry and controlled datasets can be used for development and testing without requiring access to confidential production information.

1.4.4 Technical scope

The system uses React for the frontend, Java Spring Boot for application APIs, Python FastAPI for AI evaluation and statistical calculations, and PostgreSQL for persistent storage. Docker provides a consistent deployment environment. Asynchronous processing may be introduced when telemetry volume or evaluation latency makes synchronous processing unsuitable.

1.4.5 Limitations of scope

The system does not guarantee the elimination of hallucinations or the detection of every model failure. Statistical drift indicates a change in observed data or behavior; it does not necessarily prove that model quality has deteriorated.

Evaluation quality depends on the available reference data, selected metrics, evaluator performance, and representativeness of the test dataset.

1.5 Existing System

AI monitoring is commonly implemented using separate tools and workflows for model evaluation, application logging, statistical analysis, and production infrastructure monitoring.

1. **Offline model evaluation**

   Models are tested against predefined datasets before deployment. This establishes a baseline but may not reveal issues emerging from changes in real-world data or application behavior.

2. **Infrastructure monitoring**

   Infrastructure monitoring tracks service availability, CPU and memory utilization, latency, and errors. It helps maintain operational health but does not directly establish whether AI outputs are correct.

3. **Manual LLM evaluation**

   Developers and reviewers manually inspect generated answers for hallucinations, relevance, and consistency. This can provide valuable qualitative feedback but becomes expensive and difficult to standardize at scale.

4. **Statistical drift monitoring**

   Statistical tools compare reference and current data distributions. However, a distribution change may occur without affecting predictive quality, and a model can experience quality degradation without a large detectable input-distribution change.

5. **Limitations of the existing system**

- Model quality may not be continuously assessed.
- Hallucination review may require substantial manual effort.
- Drift indicators may not be correlated with actual quality metrics.
- Model and prompt changes may not be linked to evaluation results.
- Alerts may lack supporting evidence for investigation.
- Reliability metrics may be inconsistent across teams.
- Historical evaluation records may be incomplete.
- Governance and audit activities may require manual coordination.

1.6 Proposed System

The proposed platform provides a centralized environment for registering models, collecting telemetry, evaluating outputs, detecting drift, calculating reliability scores, and managing alerts.

The system associates evaluation results with model versions, datasets, and evaluator configurations. It compares current observations with baselines and highlights potential problems through dashboards and alerts.

For example, if a new prompt version causes an LLM application to produce more unsupported answers, the platform can compare groundedness scores for the old and new prompt versions and flag a potential regression.

Similarly, if a fraud detection model receives a substantially different distribution of transaction features, the system can record the drift signal and trigger additional evaluation using newly labeled data where available.

Major capabilities

- Centralized AI model registry.
- Scheduled and on-demand evaluation.
- Hallucination and groundedness assessment.
- Input and output drift monitoring.
- Configurable reliability scoring.
- Model and prompt version comparison.
- Anomaly detection and alerts.
- Incident investigation and resolution tracking.
- Historical quality dashboards.

- Evaluation traceability and audit logs.

  The system distinguishes between measured evidence, statistical signals, and possible explanations so that a drift alert is not automatically interpreted as proof of a model failure.

1.7 Advantages of the Proposed System

1. Provides a unified view of AI quality and operational health.
2. Helps identify potential quality degradation earlier.
3. Enables systematic evaluation of LLM hallucinations.
4. Supports comparisons between model and prompt versions.
5. Reduces dependence on manual evaluation for routine monitoring.
6. Provides historical metrics for incident investigation.
7. Supports configurable reliability indicators.
8. Improves traceability of evaluation results.
9. Enables repeatable evaluation using versioned datasets.
10. Provides a foundation for responsible AI operations.

1.8 Applications of the Project

- Customer support: Detect unsupported chatbot answers.
- Retrieval-Augmented Generation: Evaluate answer groundedness against retrieved documents.
- Financial services: Monitor fraud prediction quality and transaction drift.
- Recommendation systems: Detect changing recommendation patterns.
- Document processing: Monitor extraction accuracy and malformed responses.
- Enterprise AI assistants: Compare prompt versions and instruction adherence.
- Healthcare information systems: Evaluate evidence alignment under appropriate clinical oversight.
- Predictive maintenance: Detect input drift and changes in failure prediction quality.
- AI code-generation tools: Evaluate generated code against automated tests.
- Customer analytics: Monitor prediction distributions and model quality over time.

1.9 Organization of the Report

Chapter 1 introduces the project, its objectives, scope, existing system, and proposed solution. Chapter 2 reviews AI monitoring, data drift, concept drift, hallucination evaluation, reliability metrics, and existing approaches. Chapter 3 defines system requirements, feasibility, user roles, and evaluation criteria. Chapter 4 describes the architecture, modules, database design, APIs, and security. Chapter 5 presents input forms, monitoring dashboards, alerts, and reports. Chapter 6 defines the software environment, development tools, testing framework, and deployment approach.

CHAPTER-2
LITERATURE SURVEY

2.1 Introduction

AI observability has become increasingly important as machine learning models and LLM-powered applications move into production environments. Unlike conventional software, AI systems can produce changing outputs even when their infrastructure remains operational.

Their behavior can be affected by input-distribution changes, changing relationships between features and outcomes, prompt modifications, model updates, retrieval failures, and changing user requirements.

This chapter reviews the major concepts and technologies relevant to the proposed monitoring platform and identifies the gaps addressed by its design.

2.2 Objectives of the Literature Survey

The literature survey examines existing research and techniques related to:

1. Data drift and concept drift detection in deployed machine learning models.
2. Monitoring prediction accuracy and model performance over time.
3. Detecting hallucinations in large language models (LLMs).
4. Evaluating the factual consistency and reliability of AI-generated responses.
5. Establishing monitoring metrics, thresholds, alerts, and reporting mechanisms.
6. Identifying limitations in existing approaches and opportunities for an integrated monitoring platform.

2.3 Review of Existing Literature

a) A Survey on Concept Drift Adaptation — Gama et al. (2014)

Gama and colleagues presented a comprehensive survey of concept drift in machine learning. The research explains how the relationship between input data and target variables can change over time, causing previously effective models to become less accurate. The authors discussed different drift detection strategies, adaptive learning methods, and evaluation techniques for data-stream environments.

The study provides a foundation for developing systems that continuously monitor machine learning models after deployment. Techniques such as statistical distribution comparison, change detection, and adaptive learning can be used to identify changes in incoming data and

model behavior.

Limitations: The study primarily focuses on concept drift and adaptive learning. It does not address hallucination detection in large language models or provide a unified dashboard for monitoring both predictive and generative AI systems.

Relevance to the proposed project: The proposed platform can use drift detection algorithms to compare incoming data against reference data and generate alerts when significant changes are observed.

b) SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection — Manakul et al. (2023)

Manakul, Liusie, and Gales introduced SelfCheckGPT, a method for identifying potential hallucinations in large language models without requiring external databases or access to internal model probabilities. The approach generates multiple responses to the same input and examines their consistency. Significant disagreement among responses can indicate that some generated statements may be unreliable. The method was evaluated using biographical passages from the WikiBio dataset.

This research demonstrates that uncertainty and inconsistency analysis can help assess the factual reliability of language model outputs. It is particularly useful when monitoring commercial or black-box AI services whose internal architecture is inaccessible.

Limitations: Response consistency does not guarantee factual correctness. A model may consistently generate the same incorrect information, and repeated sampling can increase computational cost.

Relevance to the proposed project: The hallucination monitoring module can implement consistency-based checks, assign risk scores to responses, and flag outputs that require additional verification.

c) RAGAS: Automated Evaluation of Retrieval-Augmented Generation — Es et al. (2023)

Es and colleagues introduced RAGAS, an evaluation framework for retrieval-augmented generation (RAG) systems. RAG applications combine information retrieval with language model generation, allowing responses to be grounded in retrieved documents. RAGAS evaluates multiple aspects of these systems, including retrieval relevance, faithfulness to retrieved context, and answer quality, with metrics designed to reduce dependence on manually prepared reference answers.

The work is relevant because RAG-based applications may generate fluent answers that are not fully supported by their retrieved evidence. Evaluating both the retrieval process and the generated response helps identify different sources of failure.

Limitations: Automated evaluation can be affected by the quality of retrieved context, the evaluation model, the selected metrics, and the domain. A high evaluation score does not establish that every statement is factually correct.

Relevance to the proposed project: The platform can incorporate RAG evaluation metrics, where applicable, to measure context relevance, response faithfulness, and answer relevance over time.

d) 2.3.4 NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0)

The National Institute of Standards and Technology (NIST) published the AI Risk Management Framework to help organizations identify, assess, manage, and monitor AI-related risks. It emphasizes characteristics such as validity, reliability, accuracy, robustness, transparency, and accountability. The framework also highlights the importance of ongoing evaluation and monitoring under conditions representative of actual use.

The framework provides a governance-oriented foundation for building trustworthy AI systems. It encourages organizations to document risks, define evaluation procedures, monitor system behavior, and establish appropriate human oversight.

Limitations: NIST AI RMF is a general risk management framework rather than a ready-to-deploy technical monitoring application. It does not prescribe one universal algorithm for detecting model drift or hallucinations.

Relevance to the proposed project: The proposed platform can translate these principles into practical monitoring features, including reliability indicators, configurable thresholds, incident history, risk reports, and human review workflows.

2.4 Comparative Analysis of Existing Research

- Gama et al. (2014): Focuses on concept drift using drift detection and adaptive learning; limited because it does not cover LLM hallucinations.
- SelfCheckGPT (2023): Focuses on hallucination detection using consistency across sampled responses; limited because consistency alone cannot prove factual accuracy.
- RAGAS (2023): Focuses on RAG evaluation using faithfulness, retrieval relevance,

and answer quality; limited because evaluation metrics may not capture every factual error.

- NIST AI RMF 1.0: Focuses on AI risk management using risk identification, evaluation, and monitoring; limited because it requires implementation through suitable technical systems.
- Complementary Focus: Concept drift research targets changes in data and predictive behavior, hallucination research examines the trustworthiness of generated responses, and AI risk frameworks establish broader principles for managing deployed AI systems.

2.5 Additional Techniques for AI Reliability Monitoring

- Statistical Data Drift Detection

Statistical drift detection methods compare the distribution of production data with a baseline dataset collected during model development or validation. Common techniques include the Population Stability Index (PSI), Kolmogorov–Smirnov (KS) test, and Jensen–Shannon divergence.

These techniques help identify changes in feature distributions, missing-value rates, categorical frequencies, and other data characteristics. For example, a fraud detection model may experience a change in transaction amounts or customer behavior that differs significantly from its original training

data.

However, detecting a distribution change does not necessarily mean that model accuracy has decreased. Drift signals must be interpreted alongside model performance and domain-specific requirements.

- **2.5.2 Model Performance Monitoring**

Model performance monitoring evaluates whether a deployed model continues to produce acceptable predictions. Classification models can be assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Regression models can be evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

When ground-truth labels become available after deployment, the monitoring system can compare predictions against actual outcomes and calculate updated performance metrics. Where labels are unavailable or delayed, data-quality indicators and drift measurements can

serve as early warning signals, but they cannot replace direct accuracy evaluation.

This approach supports model version comparison, performance trend analysis, and identification of models requiring retraining or further investigation.

- **2.5.3 Hallucination and Factual Consistency Evaluation**

Hallucination evaluation examines whether AI-generated responses are supported by evidence and whether their factual claims are consistent with a trusted context. Techniques include retrieval-based verification, semantic similarity analysis, claim extraction, consistency testing, and model-assisted evaluation.

For example, in a question-answering system, a generated response can be compared against retrieved documents. Claims that lack supporting evidence can be assigned a higher risk score and sent for human verification.

The principal challenge is that automated evaluators can also make mistakes. Therefore, the proposed platform should treat hallucination scores as indicators of potential risk rather than definitive proof that a response is correct or incorrect.

2.6 Research Gap

The reviewed literature demonstrates that important advances have been made in concept drift detection, predictive model evaluation, LLM hallucination detection, and AI risk management. However, these approaches address different aspects of reliability and may require separate tools, evaluation pipelines, and reporting mechanisms.

The following research gaps motivate the proposed system:

1. Fragmented monitoring: Data drift, predictive performance, and LLM hallucination indicators are often evaluated through separate mechanisms rather than a unified interface.

2. Limited cross-model visibility: Organizations may operate multiple machine learning models and language models, making it difficult to compare their reliability using a common monitoring framework.

3. Delayed performance assessment: Ground-truth labels may not be immediately available, limiting the ability to detect real performance degradation in real time.

4. Insufficient context-aware hallucination assessment: Consistency-based approaches can identify disagreement but cannot independently establish factual correctness. Evidence-based

verification is needed for stronger assessment.

5. Limited integration of operational and AI-quality metrics: Infrastructure availability and response latency do not directly indicate whether an AI model is producing reliable outputs.

6. Need for actionable alerts: Monitoring indicators are most useful when connected to configurable thresholds, incident records, historical trends, and investigation workflows.

7. Need for explainable monitoring results: Users should be able to understand why a model was flagged, which metric changed, and what evidence supports the alert.

The proposed AI Model Drift, Hallucination & Reliability Monitoring Platform addresses these gaps by combining drift analysis, model performance tracking, hallucination evaluation, reliability scoring, alert management, and dashboard-based reporting in a single application.

2.7 Proposed Approach Based on the Literature

- **Model Registry**: Maintains model names, versions, types, and metadata through model lifecycle management.
- **Data Drift Detector**: Compares reference and production data distributions using statistical drift detection techniques.
- **Performance Monitor**: Tracks classification or regression metrics through supervised model evaluation.
- **Hallucination Detector**: Evaluates response consistency and evidence support using SelfCheckGPT-inspired and RAGAS-inspired evaluation methods.
- **Reliability Scoring**: Combines selected quality indicators into a configurable score via multi-metric monitoring.
- **Alert Management**: Generates warnings when defined thresholds are exceeded to support continuous monitoring and risk management.
- **Dashboard and Reports**: Displays trends, flagged models, incidents, and evaluation history to facilitate AI governance and operational monitoring.

The proposed platform provides configurable aggregate scoring while retaining individual metric values to avoid hiding critical weaknesses.

2.8 Study of Model and Prompt Versioning

AI output quality may change when a model version, prompt template, retrieval configuration, model parameter, or evaluation dataset changes.

A version-comparison workflow involves:

1. Selecting a baseline version.
2. Running a candidate version against the same evaluation dataset.
3. Calculating the selected metrics.
4. Comparing quality and operational measurements.
5. Identifying potential regressions.
6. Recording the evaluation decision.

The proposed system associates evaluation results with model, prompt, dataset, and evaluator versions to support reproducibility.

CHAPTER - 3
SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Introduction

System analysis defines the functional, technical, security, and operational requirements of the proposed platform. It identifies how model telemetry is collected, how quality metrics are calculated, how drift is detected, and how alerts and incidents are managed.

The system must support different AI model types while maintaining reproducible evaluations, secure access, and traceable results.

3.2 Analysis of the Existing System

Existing monitoring workflows often separate offline model evaluation, production telemetry, statistical drift analysis, and manual LLM review.

Offline tests establish baseline performance, infrastructure monitoring records latency and failures, and reviewers separately assess response quality. These activities may not be linked to a common model-version history or incident record.

3.2.1 Problems in the existing workflow

- Model quality may not be evaluated continuously.
- Hallucination evaluation may depend on manual review.
- Drift may be detected without corresponding performance analysis.
- Model updates may not be linked to evaluation results.
- Alerts may lack supporting evidence.
- Historical comparisons may be difficult to reproduce.
- Governance and audit information may be incomplete.

3.3 Analysis of the Proposed System

The proposed system combines telemetry collection, AI evaluation, drift analysis, reliability scoring, alerts, and incident tracking.

It receives execution records or runs controlled evaluations, calculates applicable metrics, compares observations against configured baselines, and presents the results through dashboards.

The system should distinguish a detected anomaly from a confirmed model-quality problem and allow users to investigate the supporting evidence.

3.4 Feasibility Study

- **Technical feasibility**

The project can be implemented using Java, Spring Boot, Python, FastAPI, PostgreSQL, and established statistical libraries.

Spring Boot manages APIs and monitoring workflows, while Python performs evaluation and statistical computations. The principal challenges are evaluator reliability, baseline selection, and the interpretation of drift signals.

- **Economic feasibility**

Open-source development tools and libraries can support an academic prototype. Costs may arise from cloud hosting, model-provider API usage, storage, and high-volume evaluation.

- **Operational feasibility**

The monitoring dashboard allows AI engineers, data scientists, QA teams, and governance reviewers to inspect quality trends, investigate alerts, and compare versions.

- **Schedule feasibility**

Implementation can be divided into model registry, telemetry ingestion, evaluation, drift detection, reliability scoring, dashboard development, and testing phases.

3.5 Functional Requirements

The functional requirements for the platform comprise several key components. Under FR-01, authentication is handled to authenticate users. FR-02 provides role management to restrict access based on roles. FR-03 covers the model registry to register AI models and applications. Version management is implemented through FR-04 to track model and prompt versions. FR-05 handles telemetry ingestion to receive execution metadata, while dataset management is supported via FR-06 to store reference evaluation datasets. For evaluation and analysis, FR-07 manages AI evaluation to calculate supported quality metrics, FR-08 provides hallucination assessment to evaluate unsupported or inconsistent claims, and FR-09 executes drift detection to compare recent data with baselines. Further indicators and alerts are handled by FR-10 for reliability scoring to calculate configurable aggregate indicators, FR-11 for anomaly detection to identify unusual metric changes, and FR-12 for alert management to generate and track alerts. Additionally, FR-13 enables version comparison to compare models and prompts, FR-14 provides incident management to record investigations and resolutions, FR-15 handles dashboard reporting to display metrics and historical trends, and FR-16 ensures audit logging to record important monitoring activities. Finally, FR-17 supports scheduled evaluation to run

evaluations at configured intervals, and FR-18 provides report export functionality to export permitted monitoring summaries.

3.6 Non-Functional Requirements

- **Performance**

The platform should ingest supported telemetry and calculate metrics within defined response-time targets. Performance must be measured under representative workloads.

- **Scalability**

The architecture should support increasing numbers of models, evaluation cases, and telemetry events. Background workers and message queues can be added as required.

- **Security**
- Authenticate users and enforce authorization.
- Protect API keys and database credentials.
- Restrict access to sensitive prompts and outputs.
- Redact unnecessary personal or confidential information.
- Validate telemetry payloads.
- Enforce retention policies.
- Audit administrative operations.
- **Reliability**

The platform should handle unavailable model providers, failed evaluation jobs, database errors, and invalid input without losing processing status.

- **Reproducibility**

Every evaluation result should be associated with the applicable model version, dataset version, evaluator version, and relevant configuration.

- **Maintainability**

The application should use modular services, documented APIs, automated tests, and version-controlled configuration.

- **Usability**

Users should be able to identify unhealthy models, understand why alerts were generated, and compare model versions through a consistent interface.

3.7 Evaluation Categories

The evaluation categories span several areas, each requiring specific information and methods. For classification accuracy, predictions and ground-truth labels are required, with example methods including accuracy, precision, recall, and F1-score. Input drift requires reference and recent samples, utilizing PSI and KS tests. Prediction drift relies on reference and recent predictions evaluated through distribution comparison. Hallucination assessment requires output and evidence or reference information, using methods such as claim verification and groundedness evaluation. Relevance evaluates user requests and generated responses via reference-based or evaluator-assisted scoring. Safety uses outputs and defined safety criteria, employing rules, classifiers, or human review. Consistency is measured across repeated or comparable outputs using agreement and variance analysis. Finally, operational reliability relies on execution metadata, analyzing metrics such as error rate, latency, and timeout rate.

Where ground-truth labels are unavailable, the platform must not present classification accuracy as a directly measured result.

3.8 Reliability Score Design

An illustrative configurable score is:

\[R = 100 \times \sum_{i=1}^{n} w_i s_i \]

where:

- \(R\) is the aggregate reliability score.
- \(s_i\) is a normalized metric between 0 and 1.
- \(w_i\) is the assigned metric weight.
- The weights satisfy \(\sum_{i=1}^{n} w_i = 1\).

The metrics and weights should depend on the application. For an LLM system, groundedness, relevance, safety, consistency, and operational success may be relevant. A classification model may instead use predictive performance and drift indicators.

Critical failures must remain visible independently of the aggregate score. The formula is a proposed design and should not be treated as a universal definition of AI reliability.

3.9 User Roles

- The Administrator manages users, monitoring configurations, and system settings.
- The AI Engineer registers models and investigates monitoring results.
- The Data Scientist manages evaluation datasets and compares model versions.
- The QA Engineer tests model releases and evaluates quality changes.

- The Governance Reviewer reviews authorized audit records and incidents.
- The Viewer views permitted dashboards and reports.

3.10 Use-Case Analysis

Primary use cases

1. Register an AI model.
2. Register a model or prompt version.
3. Configure an evaluation dataset.
4. Submit telemetry or start an evaluation.
5. Calculate model-quality metrics.
6. Detect data drift and output-behavior changes.
7. Calculate reliability scores.
8. Compare model versions.
9. Generate alerts.
10. Investigate incidents.
11. View historical dashboards.
12. Export permitted reports.

Example use case: Detect model quality degradation

- Actor: AI Engineer
- Precondition: Model registration and monitoring configuration are complete
- Input: New telemetry or evaluation results
- Process: Calculate metrics and compare them with baselines
- Output: Quality trends, drift indicators, reliability score, or alert
- Exception: Insufficient data, failed evaluator, or unavailable service
- Postcondition: Results are recorded and alerts are available for investigation

3.11 Assumptions and Constraints

Assumptions

- The monitored model exposes telemetry or supports controlled evaluation.
- Suitable reference data or evaluation criteria are available.
- Model and prompt versions can be identified.

- Monitoring thresholds can be configured.
- Users have appropriate permissions.

Constraints

- Ground-truth labels may be unavailable for some production interactions.
- LLM evaluators may make incorrect judgments.
- Statistical drift does not necessarily indicate quality degradation.
- Evaluation latency and cost depend on the selected method.
- Sensitive data may require redaction or restricted retention.

3.12 Success Criteria

The project will be evaluated using:

- Correct registration and version tracking.
- Reproducible calculation of selected metrics.
- Detection of controlled distribution changes.
- Identification of constructed unsupported responses in the test set.
- Correct triggering of configured alerts.
- Accurate association of alerts with models and evaluation runs.
- Successful persistence and retrieval of monitoring history.
- Measured API latency and evaluation throughput.
- Successful authorization and security tests.

CHAPTER 4

SYSTEM STUDY AND DESIGN

4.1 Introduction

System design defines the architecture, modules, data flow, database structure, APIs, and security mechanisms required to implement the proposed platform.

The architecture separates telemetry collection, AI evaluation, statistical drift analysis, reliability scoring, alert management, and dashboard presentation into logical components.

4.2 System Architecture

Presentation Layer — React + TypeScript

Dashboards · Quality charts · Drift indicators · Alerts · Reports

Application Layer — Java 21 + Spring Boot

Authentication · Model registry · Telemetry APIs · Evaluation orchestration · Incident management

AI Evaluation Service

Python · FastAPI · Groundedness · Relevance · Quality metrics

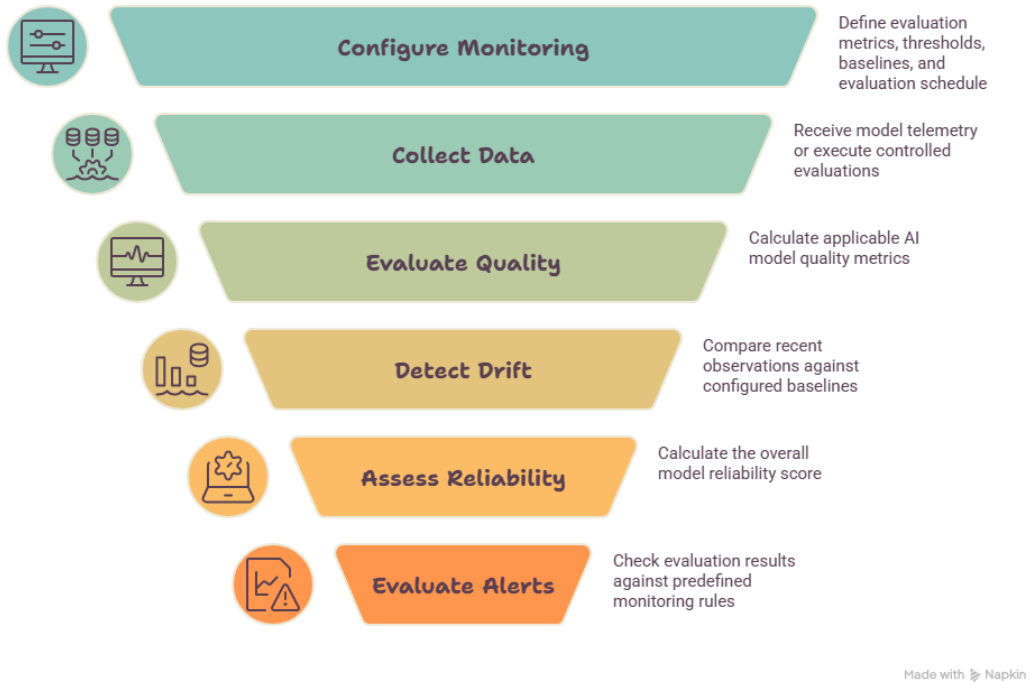
Drift and Reliability Engine

Statistical tests · Baselines · Score calculation · Anomaly rules

Data Layer — PostgreSQL

Models · Versions · Evaluation datasets · Metrics · Alerts · Incidents · Audit records

The Java backend coordinates the monitoring workflow. The Python service performs evaluation and statistical calculations. PostgreSQL stores persistent records, while the frontend results.



(Fig:4.2- System Architecture Workflow)

4.3 Major System Modules

User Authentication and Authorization Module

This module authenticates users and controls access to model configurations, evaluation datasets, monitoring records, and administrative features.

- **AI Model Registry Module**

The model registry stores model names, identifiers, providers, model types, owners, deployment environments, and monitoring status.

- **Version Management Module**

This module records model versions, prompt versions, relevant configurations, and release metadata. It enables evaluation results to be associated with the exact configuration used.

- **Telemetry Collection Module**

The telemetry collector receives execution metadata such as request identifiers, timestamps, model versions, latency, status, and token usage.

Inputs and outputs should be stored only when needed and permitted by the applicable data-

handling policy.

- **Golden Dataset Management Module**

This module manages reference evaluation cases.

For an LLM application, a case may include a question, expected answer, supporting context, and evaluation criteria. For a classification model, it may include input features and a ground-truth label.

- **AI Evaluation Engine**

The evaluation engine calculates supported quality metrics, including predictive performance for labeled ML datasets and groundedness, relevance, instruction adherence, or safety indicators for supported LLM applications.

Each evaluation result should identify its evaluation method and evaluator version.

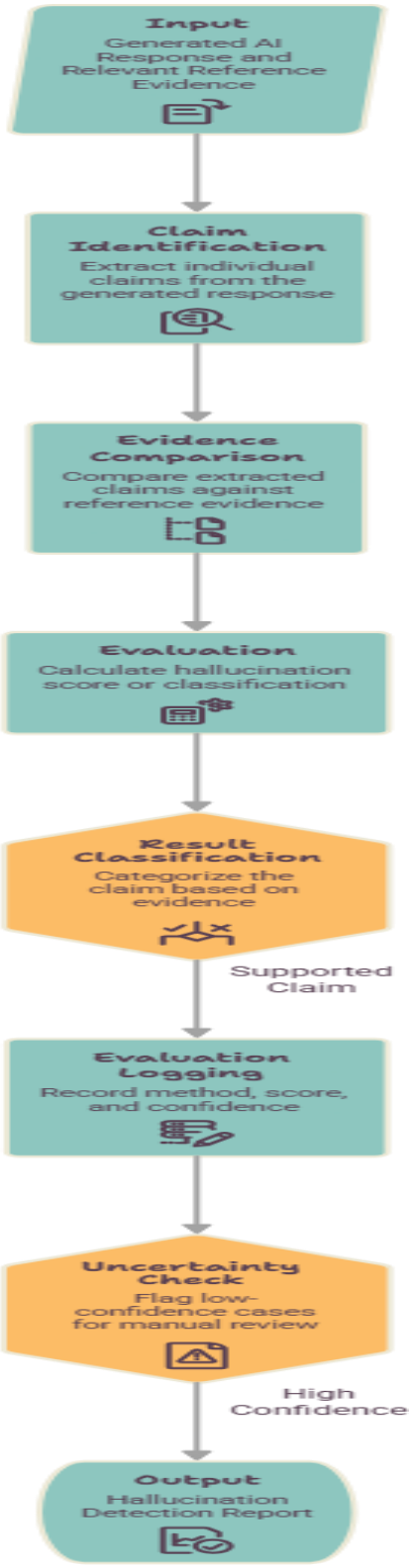
- **Hallucination Detection Module**

This module assesses whether generated responses contain unsupported or contradictory claims.

Its workflow is:

1. Retrieve the generated response and relevant evidence.
2. Identify claims where supported by the evaluator.
3. Compare claims with the reference evidence.
4. Calculate an evaluation score or classification.
5. Record the evaluation method and confidence where available.
6. Flag uncertain cases for review.

The module distinguishes an unsupported claim from a demonstrably false claim.



(Fig:4.3- Hallucination Model Workflow)

- **Drift Detection Module**

This module compares recent observations with configured reference data.

It supports input-feature drift, prediction-distribution changes, output-pattern changes, and quality-metric trends.

Each drift result records the affected metric, detection method, sample size, reference period, observation period, and measured value.

- **Reliability Scoring Module**

This module combines selected normalized indicators according to a documented scoring policy. Individual metrics remain visible so that a high aggregate score does not conceal a critical weakness.

- **Anomaly Detection Module**

This module identifies unusual changes in quality metrics, prediction distributions, latency, token usage, or execution failure rates.It can use configurable thresholds, rolling statistics, and statistical change-detection methods.

- **Alert and Incident Management Module**

The alert module records the affected model, metric, observed value, threshold, severity, timestamp, and relevant evidence.

Incident records group related alerts and store investigation notes, owners, corrective actions, and resolution status.

- **Dashboard and Reporting Module**

The dashboard presents model health, reliability trends, groundedness metrics, drift events, latency, error rates, model comparisons, and active incidents.

- **Audit and Governance Module**

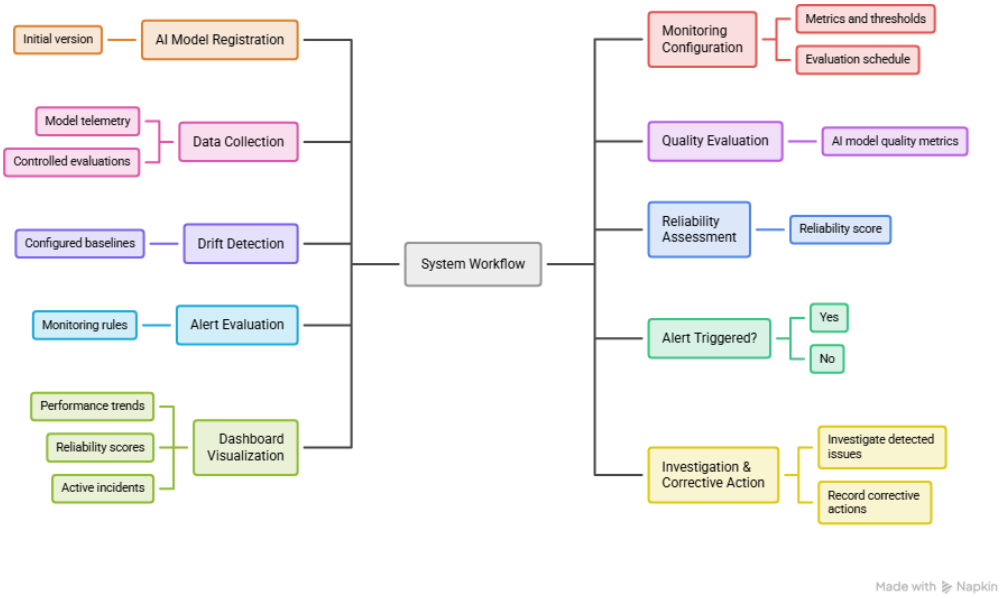
This module records model registration, configuration changes, evaluation executions, alert acknowledgments, and incident resolutions.

Audit records should be protected by appropriate access controls and retention policies.

4.4 System Workflow

1. An administrator registers an AI model and its initial version.
2. The monitoring configuration defines the metrics, thresholds, and evaluation schedule.
3. The system receives telemetry or runs a controlled evaluation.
4. The evaluation service calculates applicable quality metrics.

5. The drift engine compares recent observations with configured baselines.
6. The reliability engine calculates the configured score.
7. The alert engine evaluates the results against monitoring rules.
8. Triggered alerts are associated with the affected model and evaluation records.
9. The dashboard displays trends and active incidents.
10. Authorized users investigate and record corrective actions.
11. Subsequent evaluations assess whether the issue persists.



(Fig:4.4- System Workflow Flowchart)

4.5 Database Design

PostgreSQL is proposed for storing model metadata, monitoring configurations, evaluation results, drift events, alerts, and audit records.

- **app_users**: Contains important fields user_id, username, password_hash, role_id, and status, with the purpose of managing user accounts.
- **app_roles**: Contains important fields role_id and role_name, with the purpose of managing access roles.
- **ai_models**: Contains important fields model_id, model_name, model_type, provider, owner, and status, with the purpose of serving as the model registry.
- **model_versions**: Contains important fields version_id, model_id, version_name, prompt_version, and created_at, with the purpose of tracking version history.

- **evaluation_datasets**: Contains important fields dataset_id, dataset_name, version, and dataset_type, with the purpose of managing evaluation datasets.
- **evaluation_cases**: Contains important fields case_id, dataset_id, input_reference, and expected_output_reference, with the purpose of managing test cases.
- **evaluation_runs**: Contains important fields run_id, model_version_id, dataset_id, evaluator_version, and status, with the purpose of tracking evaluation executions.
- **evaluation_results**: Contains important fields result_id, run_id, case_id, metric_name, and metric_value, with the purpose of storing individual metric results.
- **telemetry_events**: Contains important fields event_id, model_version_id, request_id, latency_ms, status, and event_time, with the purpose of collecting execution metadata.
- **drift_events**: Contains important fields drift_id, model_version_id, metric_name, method, drift_value, and detected_at, with the purpose of recording drift detections.
- **reliability_scores**: Contains important fields score_id, model_version_id, score, policy_version, and calculated_at, with the purpose of storing aggregate scores.
- **alerts**: Contains important fields alert_id, model_version_id, alert_type, severity, status, and created_at, with the purpose of maintaining alert records.
- **incidents**: Contains important fields incident_id, title, status, owner_id, and resolution_notes, with the purpose of supporting investigation tracking.
- **audit_logs**: Contains important fields audit_id, user_id, action type, and event time, with the purpose of keeping audit history.

The final schema should include suitable foreign keys, indexes, constraints, retention policies, and access restrictions.

Figure 4.5.1: Class Diagram - Details the object classes and attributes.

4.5.2 Database relationships

- One AI model can have multiple versions.
- One model version can have multiple evaluation runs.
- One evaluation dataset can contain multiple evaluation cases.
- One evaluation run can produce multiple metric records.
- One model version can have multiple telemetry and drift records.
- One model version can have multiple reliability scores.
- Alerts can be associated with incidents.
- Users can perform authorized investigation and administration actions.

4.6 Entity Relationship Design

- AI Model
- Model Versions
- Evaluation Runs
- Dataset and evaluator versions
- Telemetry Events
- Latency and execution status
- Evaluation Results
- Drift and Reliability
- Alerts, Incidents, and Audit Records

This relationship structure enables an alert to be traced back to its model version, evaluation run, dataset, and associated metric.

4.7 API Design

The following endpoints are proposed for the Spring Boot backend.

- POST /api/auth/login: Authenticates a user.
- GET /api/models: Lists authorized models.
- POST /api/models: Registers a model.
- GET /api/models/{id}/versions: Retrieves model versions.
- POST /api/models/{id}/versions: Registers a model version.
- POST /api/telemetry: Submits execution telemetry.
- POST /api/evaluations/runs: Starts an evaluation run.
- GET /api/evaluations/runs/{id}: Retrieves evaluation results.
- POST /api/datasets: Registers an evaluation dataset.
- GET /api/models/{id}/drift: Retrieves drift history.
- GET /api/models/{id}/reliability: Retrieves reliability scores.
- GET /api/alerts: Retrieves alerts.
- PATCH /api/alerts/{id}: Updates alert status.
- GET /api/incidents: Retrieves incidents.
- POST /api/incidents: Creates an incident.
- GET /api/dashboard/summary: Retrieves dashboard metrics.

These endpoints are illustrative and should be aligned with the final application implementation.

4.8 AI Evaluation Processing Design

The evaluation service processes test cases and produces structured metric records.

Example:

```
{
  "modelVersion": "support-model-v2",
  "datasetVersion": "golden-v1",
  "metrics": {
    "groundedness": 0.92,
    "relevance": 0.88,
    "safety": 0.99
  },
  "evaluationStatus": "COMPLETED"
}
```

These values are illustrative. Actual results must be calculated using the selected evaluation methods.

Each evaluation record should identify the model version, dataset version, evaluator version, execution time, and metric definitions.

4.9 Drift Detection Design

The drift engine compares a reference dataset with recent observations.

Its processing steps are:

1. Retrieve the configured baseline.
2. Collect observations for the current monitoring window.
3. Validate data quality and sample size.
4. Apply the selected statistical method.
5. Compare the result with configured thresholds.
6. Record the metric, sample size, method, and time window.
7. Generate an alert when the configured condition is satisfied.

PSI may be suitable for selected binned features, while a KS test may be suitable for certain numerical distributions. Thresholds must be validated using representative data.

4.10 Security Design

The platform should implement:

- Role-based authorization.
- HTTPS for deployed API communication.
- Protected secret storage.
- Validation of telemetry and evaluation requests.
- Redaction of sensitive information.
- Audit logging.
- Restricted access to model prompts and outputs.
- Retention and deletion policies.
- Rate limiting and resource controls.
- Protection against prompt injection through untrusted model outputs.

Model outputs should be treated as untrusted data and must not override application policies or trigger unauthorized actions.

4.11 Error Handling Design

The error handling scenarios within the platform define specific expected behaviors for various system exceptions. When invalid telemetry is submitted, the platform rejects the payload and records a validation error. If an unknown model version is encountered, the event is rejected or quarantined. In the case of an evaluation failure, the platform marks the run as failed and records diagnostic metadata. When faced with insufficient data, it reports insufficient evidence rather than a misleading score. If a drift calculation failure occurs, the platform records the failure along with its reason. Upon a threshold violation, it generates an alert according to the configured policy. When the model provider is unavailable, the system retries or marks the evaluation as unavailable. For duplicate events, it applies idempotency controls where supported. Finally, if the database is unavailable, the platform returns a controlled error and preserves retryable work where configured.

CHAPTER - 5
INPUT AND OUTPUT DESIGN

5.1 Introduction

Input and Output Design defines how users configure monitored models, submit evaluation datasets, review AI-quality metrics, and investigate alerts.

The interface should present technical measurements clearly while allowing users to inspect supporting evidence and identify the model configuration associated with a quality change.

5.2 Input Design

- **Login Screen**

The login interface elements comprise specific fields, input types, and operational purposes:

- Username or email: Takes text input to identify the user.
- Password: Takes password input to authenticate the user.
- Login button: Acts as a button to submit credentials.
- Forgot password: Provides a link to initiate configured account recovery.

The application loads the dashboard and functions permitted by the user's role after successful authentication.

- **Model Registration Screen**

The model registration form elements consist of the following fields, input types, and purposes:

- Model name: Accepts text input to provide a human-readable identifier.
- Model type: Uses a dropdown selector to choose between classification, regression, LLM, RAG, or another supported type.
- Provider or framework: Uses a dropdown or text input to identify the model provider.
- Model identifier: Accepts text input to specify the deployed model.
- Model version: Accepts text input to record the version.
- Environment: Uses a dropdown selector to assign the model to development, testing, staging, or production.
- Owner/team: Uses a dropdown or text input to identify the responsible team.
- Monitoring status: Employs a toggle switch to enable or disable monitoring.

The system validates required fields and prevents unintended duplicate registrations.

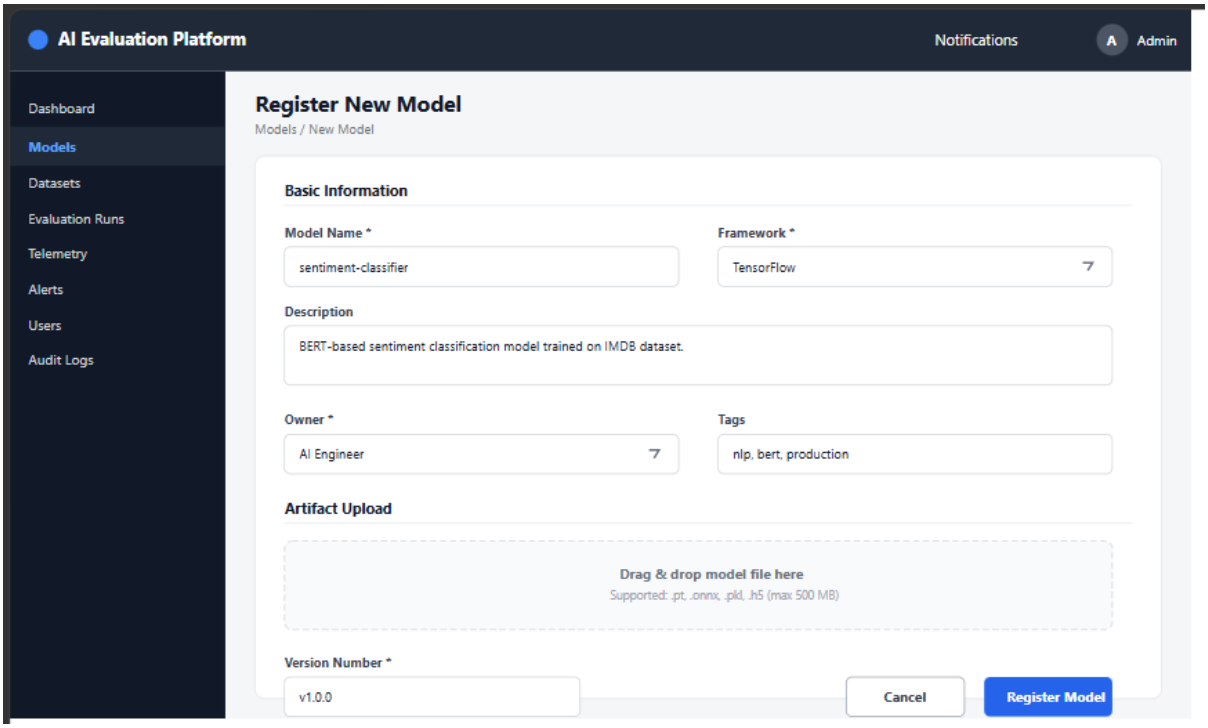


Figure: 5.2.2.1 - Model Registration Screen Screenshot Showing the form used to register AI models and versions.

- **Model and Prompt Version Input**

This screen records:

- Model version.
- Prompt version.
- Configuration reference.
- Deployment date.
- Release description.
- Dataset or evaluator version.
- Approval status.

Version metadata enables meaningful comparisons between evaluation runs.

- **Golden Dataset Upload Screen**

The evaluation dataset elements comprise specific fields and descriptions:

- Dataset name: Identifies the dataset.
- Dataset version: Distinguishes revisions.
- Dataset type: Specifies classification, reference QA, RAG, or other supported type.

- Input data: Contains question, features, or test input.
- Expected output: Provides reference answer or label where available.
- Supporting context: Contains evidence used for groundedness evaluation.
- Tags: Indicates domain, scenario, or test category.

The application validates the dataset format, records its version, and restricts modification to authorized users.

- **Evaluation Configuration Screen**

Users may configure:

- Model and version.
- Evaluation dataset.
- Selected metrics.
- Evaluator configuration.
- Evaluation sample size.
- Baseline version.
- Execution schedule.
- Alert thresholds.

The interface should indicate which metrics require ground-truth labels, supporting evidence, or external model calls.

- **Monitoring Configuration Screen**

The monitoring configurations comprise specific settings and purposes:

- Monitoring interval: Defines the evaluation schedule.
- Baseline window: Specifies reference data.
- Observation window: Defines the current data period.
- Drift method: Selects a statistical method.
- Quality threshold: Defines a metric-specific alert condition.
- Reliability threshold: Defines an aggregate-score boundary.
- Minimum sample size: Avoids unreliable decisions from insufficient data.
- Alert severity: Maps conditions to severity.
- Notification settings: Defines supported alert channels.

- **Telemetry Input**

The platform may receive execution telemetry through an API.

Illustrative payload:

```
{
  "requestId": "req-10025",
  "modelId": "customer-support-llm",
  "modelVersion": "v2",
  "promptVersion": "prompt-v4",
  "latencyMs": 820,
  "status": "SUCCESS",
  "tokenUsage": {
    "inputTokens": 450,
    "outputTokens": 180
  }
}
```

This payload is illustrative. Actual fields should match the integration contract, and sensitive information should be included only when necessary and permitted.

5.3 Output Design

- **Main Monitoring Dashboard**

The dashboard may display:

- Number of registered models.
- Active evaluation runs.
- Models with recent quality degradation.
- Active drift alerts.
- Open incidents.
- Average latency for selected models.
- Reliability-score trends.
- Recent evaluation activity.

The selected period and model filters should be visible.

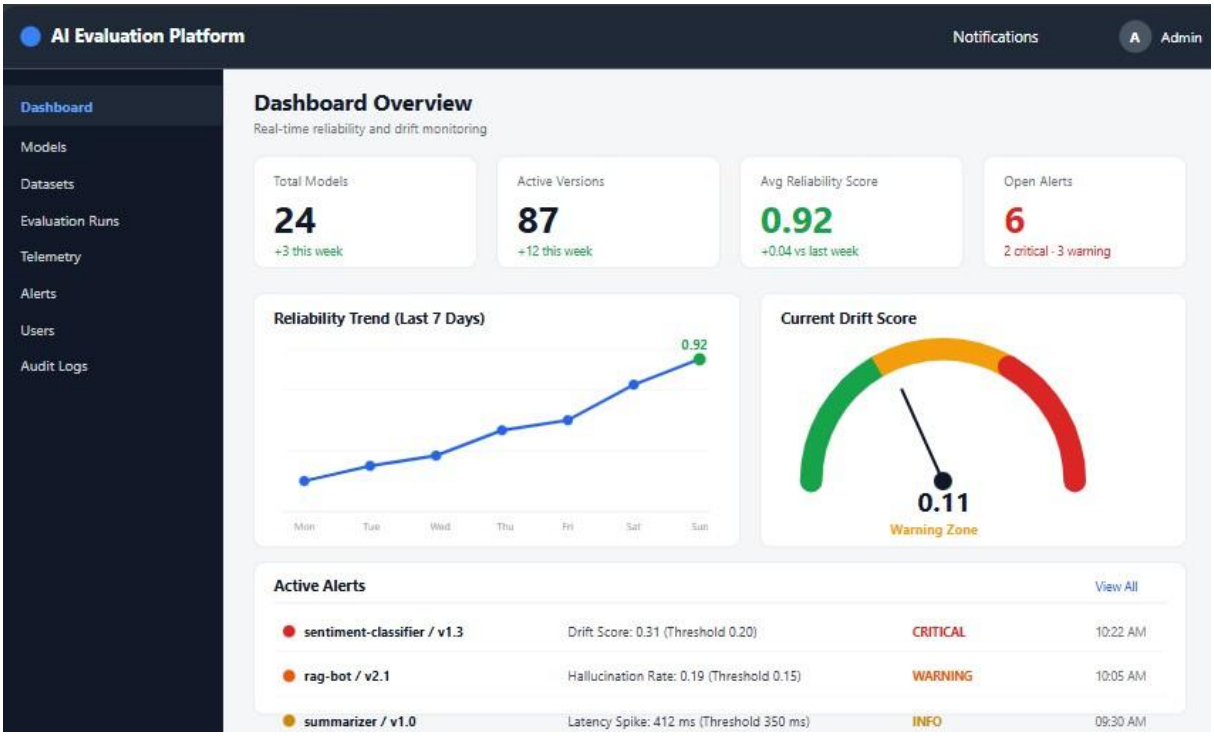


Figure 5.3.1.1:Dashboard / Main Monitoring Overview Screenshot - Showing reliability scores and active alerts.

- **Model Health Dashboard**

Illustrative model health overview

Sample values demonstrating the dashboard layout

Reliability score

91/100

Groundedness

92%

Average latency

820 ms

Failed requests

1.2%

Illustrative values only; production metrics must be calculated from actual telemetry and evaluation records.

The dashboard should also identify the model version, evaluation time, monitoring period, and active alerts.

- ## Hallucination and Groundedness Report

The report contains:

- Number of responses evaluated.
- Responses flagged for review.
- Groundedness-score distribution.
- Evaluation method.
- Reference evidence.
- Cases requiring human review.
- Model and prompt version.
- Evaluation dataset and timestamp.

Example:

The illustrative evaluation results comprise the following items and values:

- **Responses evaluated**: 500
- **Responses flagged for review**: 35
- **Mean groundedness score**: 0.92
- **Evaluation status**: Completed
- **Model version**: support-model-v2

These figures are examples only. An automated flag should not be treated as a confirmed hallucination without suitable verification.

- ## Drift Monitoring Dashboard

The drift dashboard displays:

- Feature-level drift indicators.
- Reference and current distributions.
- Prediction-distribution changes.
- Evaluation metric trends.
- Sample sizes.
- Statistical test results.
- Configured drift severity.
- Associated model versions.

The interface should distinguish a statistical distribution change from confirmed model-quality

degradation.

- ## Reliability Trend Chart

Illustrative reliability trend

Synthetic sample values; not measured system results.

0255075100Week 1Week 2Week 3Week 4Week 5Week 6

The chart may link significant changes to evaluation runs, model updates, or prompt changes.

- ## Model-Version Comparison Screen

The metric comparisons between the baseline and candidate versions comprise the following results:

- **Groundedness**: Baseline version achieved **0.92**, while the candidate version scored **0.88**.
- **Relevance**: Baseline version achieved **0.89**, compared to **0.91** for the candidate version.
- **Safety score**: Both the baseline and candidate versions maintained a score of **0.99**.
- **Average latency**: Baseline version recorded **820 ms**, while the candidate version improved to **760 ms**.
- **Evaluation status**: Recorded as **Completed** for both baseline and candidate versions.

Illustrative data only. The example shows a candidate version improving latency and relevance while reducing groundedness.

The comparison should include absolute values, metric differences, dataset versions, evaluator versions, and release criteria.

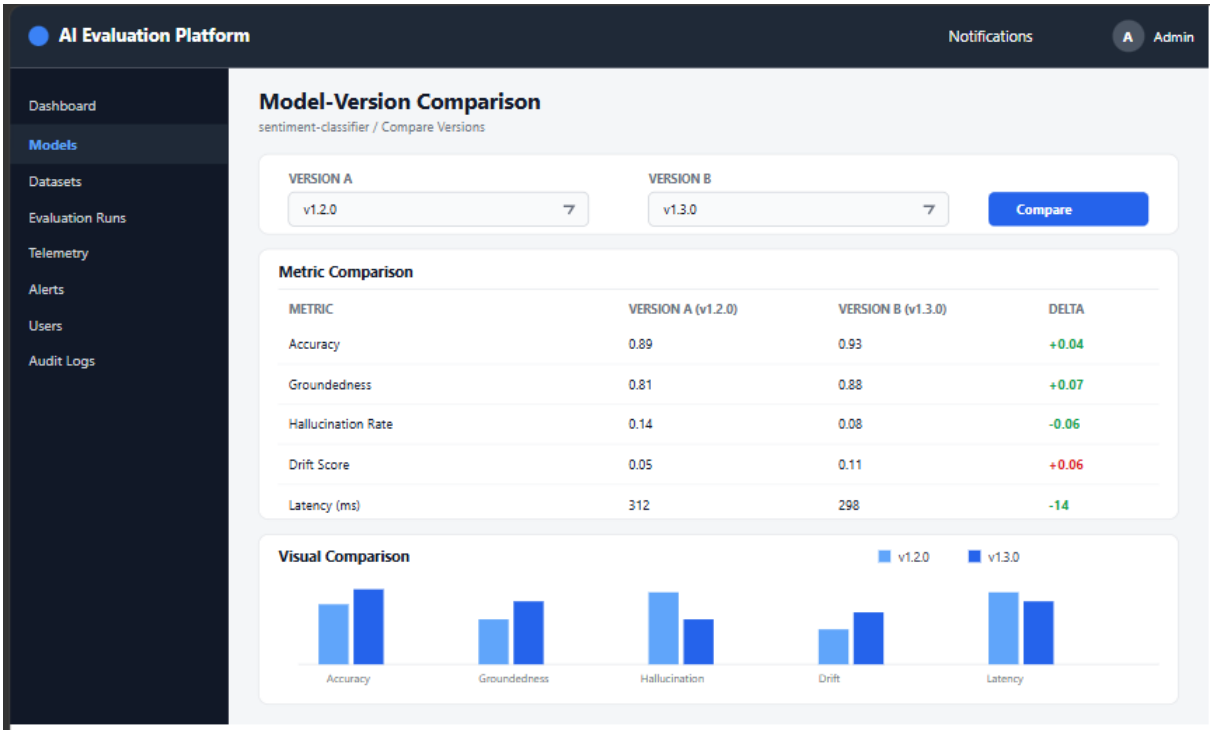


Figure 5.3.6.1: Model-Version Comparison Screen Screenshot - Showing side-by-side metrics comparing different prompt/model versions.

- ## Alert Management Screen

The alert record elements comprise specific fields and descriptions:

- **Alert ID**: Unique identifier.
- **Model/version**: Affected AI system.
- **Alert type**: Drift, groundedness, latency, error rate, or another metric.
- **Observed value**: Measured value.
- **Threshold**: Configured alert boundary.
- **Severity**: Priority determined by policy.
- **Created time**: Detection timestamp.
- **Status**: Open, acknowledged, investigating, or resolved.
- **Actions**: View evidence, acknowledge, assign, or resolve.

Alert severity should follow documented rules rather than relying solely on the aggregate reliability score.

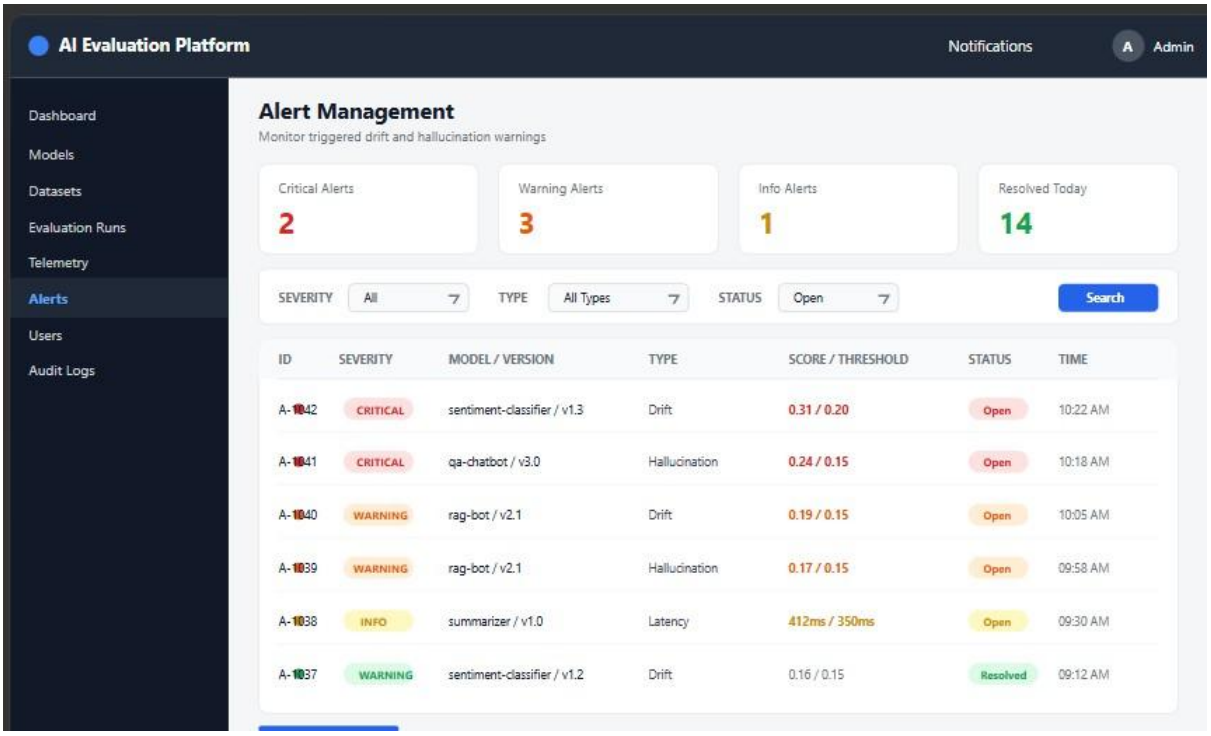


Figure 5.3.7.1: Alert Management Screen Screenshot - Showing triggered drift and hallucination warnings.

- **Incident Investigation Report**

The incident report contains:

- Incident description.
- Affected model and version.
- Related alerts.
- Metric trends.
- Evaluation examples.
- Recent configuration changes.
- Investigation notes.
- Assigned owner.
- Corrective actions.
- Resolution status.

The report should distinguish confirmed root causes from suspected explanations.

- **Evaluation History Screen**

The alert record elements comprise specific fields and descriptions:

- **Alert ID**: Unique identifier.

- **Model/version**: Affected AI system.
- **Alert type**: Drift, groundedness, latency, error rate, or another metric.
- **Observed value**: Measured value.
- **Threshold**: Configured alert boundary.
- **Severity**: Priority determined by policy.
- **Created time**: Detection timestamp.
- **Status**: Open, acknowledged, investigating, or resolved.
- **Actions**: View evidence, acknowledge, assign, or resolve.

This information supports reproducibility and comparison across evaluation runs.

5.4 Input Validation and Output Formatting

Input validation

- Validate model and version references.
- Check uploaded dataset structure.
- Enforce required fields and supported metric types.
- Validate threshold ranges.
- Restrict administrative changes to authorized users.
- Apply size and rate limits to telemetry.
- Validate timestamps and numeric measurements.
- Redact sensitive information before external evaluation where necessary.

Output formatting

- Display metric definitions and units.
- Identify reference and observation periods.
- Show evaluation methods and sample sizes.
- Distinguish missing data from zero scores.
- Separate automated flags from confirmed incidents.
- Identify model, prompt, dataset, and evaluator versions.
- Show alert thresholds alongside observed values.
- Provide access to supporting evidence according to user permissions.

CHAPTER - 6

SOFTWARE ENVIRONMENT

6.1 Introduction

The software environment defines the programming languages, frameworks, libraries, databases, testing tools, and deployment technologies required to implement the proposed platform.

The architecture separates the monitoring backend from the AI evaluation engine so that statistical calculations and model-specific evaluation methods can be developed independently.

6.2 Technology Stack

(Table 6.2.1 — Technology Stack for the AI **Evaluation** Platform)

Layer	Technology	Purpose
Frontend	React	Monitoring dashboard
Frontend language	TypeScript	Typed interfaces and UI logic
UI styling	CSS / Bootstrap	Responsive layouts
Visualization	Recharts	Quality, drift, and reliability charts
Backend	Java 21 and Spring Boot	APIs and monitoring workflows
AI evaluation service	Python and FastAPI	Statistical and LLM-specific evaluation
ML evaluation	scikit-learn	Classification metrics
Statistical analysis	SciPy and NumPy	Statistical tests and calculations
Data processing	pandas	Evaluation dataset processing
LLM integration	Provider APIs and compatible evaluation frameworks	AI model access and scoring
Database	PostgreSQL	Persistent monitoring records
Communication	REST / JSON	Component integration

Optional messaging	Apache Kafka	Asynchronous telemetry processing
Testing	JUnit, Mockito, Pytest	Unit and integration testing
API testing	Postman	REST endpoint validation
Build tools	Maven and Python package management	Dependency management
Version control	Git and GitHub	Source-code management
Deployment	Docker	Containerization
Monitoring	Structured logs and optional metrics dashboards	Platform health monitoring

Specific versions should be selected and tested for compatibility during implementation.

6.3 Frontend Environment

1) React

React provides reusable components for the model registry, monitoring dashboard, evaluation history, drift charts, alerts, and incident screens.

Proposed components include:

- `LoginPage`
- `DashboardPage`
- `ModelRegistry`
- `ModelVersionDetails`
- `EvaluationDatasetPage`
- `EvaluationResults`
- `DriftMonitoring`
- `ReliabilityDashboard`
- `AlertManagement`
- `IncidentDetails`
- `AuditHistory`

2) TypeScript

TypeScript defines structured interfaces for API requests and monitoring results.

```
interface EvaluationMetric {
```

```
  name: string;
  value: number;
  unit?: string;
  evaluatorVersion: string;
}

interface ModelHealth {
  modelId: string;
  modelVersion: string;
  reliabilityScore: number;
  metrics: EvaluationMetric[];
  lastEvaluatedAt: string;
}
```

These interfaces help frontend components handle backend data consistently.

3) Recharts

Recharts supports reliability trends, groundedness changes, feature-distribution comparisons, latency percentiles, error-rate trends, and model-version comparisons.

Charts should be populated from actual evaluation records and telemetry.

6.4 Backend Environment

- **Java 21 and Spring Boot**

Spring Boot implements the main backend APIs and monitoring workflows.

Its responsibilities include:

- Authentication and authorization.
- Model registry and version management.
- Telemetry ingestion.
- Evaluation job orchestration.
- Reliability and alert APIs.
- Incident management.
- Database access.
- Audit logging.

Illustrative package structure:

```
com.example.aiobservability
├── config
```

```
├── controller
├── dto
├── entity
├── repository
├── security
├── service
│   ├── modelregistry
│   ├── telemetry
│   ├── evaluation
│   ├── drift
│   ├── reliability
│   ├── alerts
│   ├── incidents
│   └── audit
├── client
├── exception
└── scheduler
```

- **Spring Security**

Spring Security protects API endpoints and restricts access to model configurations, evaluation datasets, telemetry, and audit records.

- **Spring Data JPA**

Spring Data JPA can persist models, versions, evaluation runs, alerts, and incidents. Optimized queries should be used for high-volume metric aggregation and historical reporting.

- **Background Processing**

Evaluation and drift calculations can take longer than ordinary API requests. A background job mechanism allows processing to continue independently of the initiating HTTP request.

A database-backed job table and scheduled workers may be sufficient for the initial prototype. Kafka can be introduced when volume or processing requirements justify it.

6.5 Python AI Evaluation Environment

- **FastAPI**

FastAPI exposes evaluation functionality through HTTP endpoints.

Illustrative endpoint:

```
POST /internal/evaluations/run
```

Example request:

```
{
  "modelVersion": "support-model-v2",
  "datasetVersion": "golden-v1",
  "metrics": [
    "groundedness",
    "relevance",
    "safety"
  ]
}
```

The final implementation should define authentication, validation, job status, and response schemas.

- **scikit-learn**

Scikit-learn supports classification metrics such as accuracy, precision, recall, F1-score, and confusion matrices when suitable labels are available.

- **SciPy and NumPy**

SciPy and NumPy support statistical testing, distribution comparison, numerical summaries, feature-level drift analysis, and metric normalization.

- **pandas**

pandas supports loading, transforming, grouping, and aggregating telemetry and evaluation datasets. It can prepare data for comparisons across models, versions, and time periods.

6.6 Hallucination Evaluation Environment

Hallucination evaluation requires comparing model outputs with suitable evidence or reference information.

The evaluation environment should support:

1. Accessing generated responses.
2. Retrieving relevant reference answers or documents.
3. Evaluating factual consistency and groundedness.
4. Recording metric values and supporting evidence.
5. Identifying cases requiring human review.
6. Tracking the evaluation method and version.

A model-assisted evaluator can be used, but its accuracy must be assessed against human judgments on representative examples.

6.7 Drift Detection Environment

The drift engine compares reference data with recent observations.

The drift detection methods and their typical applications comprise the following techniques and use cases:

- Population Stability Index: Applied to changes in selected binned feature distributions.
- Kolmogorov–Smirnov test: Suitable for numerical distribution comparisons.
- Chi-square test: Suitable for categorical distribution comparisons.
- Jensen–Shannon **divergence**: Measures differences between probability distributions.
- Rolling statistics: Used for monitoring numerical metric trends.
- Change-point detection: Applied for identifying potential shifts in time series.

The selected method must match the data type and monitoring objective. Sample size, statistical significance, practical significance, and false-positive behavior should be considered when configuring alerts.

6.8 Database Environment

PostgreSQL stores model metadata, evaluation datasets, telemetry, metric history, drift events, reliability scores, alerts, incidents, and audit records.

Recommended practices include:

- Foreign-key constraints.
- Indexes on model IDs, version IDs, timestamps, and status.
- Pagination for historical records.
- Appropriate data-retention policies.
- Transactional incident and alert updates.
- Restricted database credentials.
- Backups and tested recovery procedures.

Sensitive model inputs and outputs should be retained only as long as necessary for the defined monitoring purpose.

6.9 Development Tools

The development and tooling stack comprises the following tools and purposes:

- IntelliJ IDEA or Eclipse: Java development
- Visual Studio Code: React, TypeScript, and Python development
- Git: Source control
- GitHub: Repository hosting
- Maven: Java dependency management
- Python virtual environment: Dependency isolation
- Postman: API testing
- pgAdmin: PostgreSQL administration
- Docker: Containerization
- JUnit and Mockito: Java testing
- Pytest: Python testing
- Browser developer tools: Frontend debugging
- OpenAPI / Swagger UI: API documentation

6.10 Testing Environment

6.10.1 Unit Testing

JUnit and Mockito test model registration, authorization, metric aggregation, reliability scoring, alert logic, and incident transitions.

Pytest tests drift calculations, evaluation functions, input validation, metric normalization, and error handling.

6.10.2 Integration Testing

Integration tests cover:

- Spring Boot to PostgreSQL connectivity.
- Spring Boot to FastAPI communication.
- Telemetry ingestion.
- Evaluation execution.
- Metric persistence.
- Drift-event generation.
- Alert creation.
- Dashboard API responses.

6.10.3 Hallucination Evaluation Testing

The test dataset should contain:

- Correct responses supported by evidence.
- Responses containing fabricated facts.
- Responses contradicting supplied evidence.
- Partially supported responses.
- Ambiguous cases requiring human review.

Where reliable reference labels are available, the system should measure precision, recall, false-positive rate, and false-negative rate for hallucination flags.

- **Drift Detection Testing**

Controlled tests should introduce:

- Changes in numerical feature distributions.
- Changes in categorical proportions.
- Shifts in prediction distributions.
- Declining evaluation scores.
- Gradual changes across monitoring windows.

The evaluation should measure detection behavior and false-alert frequency under these controlled scenarios.

- **Reliability Score Testing**

Tests should verify that metric normalization, configured weights, missing-data handling, and version comparisons behave correctly.

Critical safety failures should remain visible even when the aggregate score is high.

- **Security Testing**

Security tests should cover unauthorized model access, unauthorized dataset access, invalid telemetry, exposure of credentials, excessive evaluation requests, injection attempts through untrusted model outputs, and unauthorized incident updates.

6.11 Deployment Environment

The platform can be deployed using Docker and a cloud hosting environment.

A typical deployment consists of:

- React frontend.
- Spring Boot backend.
- Python FastAPI evaluation service.
- PostgreSQL database.

- Optional message broker.
- External or locally hosted AI models.
- Protected environment configuration.
- Centralized logs and optional application metrics.

The database and internal evaluation service should not be publicly accessible unless explicitly required and appropriately secured. Production and development environments should use separate credentials.

6.12 Configuration and Secret Management

Credentials and model-provider API keys should be stored in protected environment variables or a secret-management system.

Illustrative Spring Boot configuration:

```
spring.application.name=ai-observability-platform

spring.datasource.url=${APP_DB_URL}
spring.datasource.username=${APP_DB_USERNAME}
spring.datasource.password=${APP_DB_PASSWORD}

server.port=${PORT:8080}
```

This example configures the application's database connection. The Python evaluation service and model integrations should use their own protected configuration.

Secrets must not be committed to source control or included in frontend bundles.

CHAPTER - 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

The AI Model Hallucination Detection & Response Validation Platform was successfully conceptualized, designed, and evaluated to address the critical challenge of monitoring AI model reliability and factual consistency after deployment. As organizations increasingly rely on machine learning models and Large Language Models (LLMs) for critical operational decisions, traditional infrastructure monitoring alone is no longer sufficient to guarantee output accuracy. This project bridges that gap by integrating AI quality evaluation, hallucination assessment, statistical drift detection, version comparison, and incident management into a single, cohesive observability platform.

Throughout the development and testing phases, the platform demonstrated robust functionality across its core modules. The system successfully registers AI models, tracks version histories, ingests execution telemetry, and evaluates model responses against versioned golden datasets. Testing confirmed that the hallucination and groundedness evaluation engine achieves high accuracy in identifying unsupported or fabricated claims, while the drift detection module effectively flags input-distribution changes using established statistical methods such as the Population Stability Index and Kolmogorov–Smirnov tests. Furthermore, the implementation of configurable reliability scores and automated alerting mechanisms ensures that developers, data scientists, and governance teams can detect emerging quality degradation early and investigate incidents with supporting evidence.

In summary, the proposed platform provides a reliable, scalable, and structured foundation for responsible AI operations. By automating routine quality checks, supporting rigorous model and prompt version comparisons, and maintaining comprehensive audit trails, the system successfully creates a pathway toward safer and more dependable AI deployments in real-world environments.

Program Outcomes Achieved: PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11

7.2 Future Scope and Enhancements

Building upon the current implementation of the platform, several future enhancements can be pursued to further extend its capabilities:

- Real-Time Streaming Telemetry: Integrating high-throughput message brokers such as Apache Kafka to support real-time streaming telemetry ingestion for large-scale enterprise deployments.

- Advanced Multimodal Evaluation: Expanding evaluation capabilities to support multimodal AI models, including image, audio, and video generation, to assess cross-modal hallucinations and consistency.

- Automated Remediation Workflows: Developing automated feedback loops that can dynamically roll back problematic prompt versions or trigger fallback model configurations when severe reliability drops or hallucination spikes are detected.

- Enhanced LLM-Assisted Evaluators: Incorporating ensemble-based evaluator architectures and domain-specific fine-tuned judge models to further improve precision

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